BLUESTOCK FINTECH

Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Business Intelligence Dashboard

Prepared For: Bluestock Fintech Pvt. Ltd.
Prepared By: Intern / Data Analyst — Bluestock Fintech
Project Domain: Wealth Management / WealthTech Analytics
Version: 1.0 (Production Release)
Date: June 2026

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1. Executive Summary

1.1 Business Context & Scope

The Indian Mutual Fund industry is undergoing massive growth, driven by widening financial literacy and expanding equity participation among retail and corporate investors. As of December 2025, the industry manages over Rs. 81 lakh crore in Assets Under Management (AUM) across 1,908 schemes. Monthly SIP inflows have breached Rs. 31,000 crore, reflecting the deepening equity culture. Despite this expansion, individual investors and corporate financial advisers face significant challenges in selecting optimal mutual fund schemes. Fragmented data across multiple websites, complex volatility metrics (Sharpe, Sortino, VaR), and opaque benchmark tracking create significant decision-making friction.

Bluestock Fintech has commissioned this project to design and build an end-to-end, production-grade Mutual Fund Analytics Platform. The scope spans daily historical NAV ingestion (40 key schemes, Jan 2022 to May 2026), quarterly AUM monitoring (top 10 fund houses), demographic transaction segmentation (32,000+ transaction logs of 5,000 investors across 12 states), index-linked alpha calculations, and sector portfolio concentration HHI. The platform aims to democratize wealth analytics by delivering institutional-grade indicators in a clear, self-service dashboard.

1.2 Strategic Platform Vision

By integrating data engineering pipeline efficiency with predictive statistical intelligence, this platform serves as a unified tool. For wealth distributors and wealth managers, the platform reduces research cycles from days to seconds. For retail users, it demystifies quantitative risk by visualising return-to-risk efficiency, historical maximum drawdown, and portfolio sector diversification.

1. Executive Summary (Continued)

1.3 Core Project Accomplishments

Over the 7-day development lifecycle, the following modules were fully built, integrated, and verified:

- **Phase 1 Ingestion:** Built daily NAV fetchers targeting public APIs (mfapi.in), recovering ~46,000 aligned records anchored to real-world AMFI values.
- **Phase 2 Database DDL:** Formulated a normalized star schema database using SQLite with 8 dimension and fact tables (dim_fund, dim_date, fact_nav, fact_transactions, fact_performance, fact_portfolio, fact_aum, fact_sip_industry) optimized with indexes for query execution speeds under 5ms.
- **Phase 3 ETL Preprocessing:** Coded robust cleaning filters that resolve weekends/market holidays using forward-filling (ffill), normalize string nomenclature, drop duplicates, and enforce schema constraint checks.
- **Phase 4 Performance Calculator:** Computed 1-year, 3-year, and full-period CAGRs, Sharpe and Sortino ratios (Rf = 6.5%), OLS index-regression (Alpha, Beta, R-squared), and maximum peak-to-trough drawdowns.
- **Phase 5 Advanced Analytics:** Coded historical Value at Risk (95% VaR) and Conditional VaR (95% CVaR), Herfindahl-Hirschman Index (HHI) for sector concentration, and investor transaction cohort and SIP continuation metrics.
- **Phase 6 BI Dashboard:** Designed a 4-page interactive Power BI dashboard featuring dynamic AMC filter slicers, risk-return scatter visualizations, and geographic distribution heatmaps.
- **Phase 7 Orchestration:** Designed a master execution script (run_pipeline.py) allowing single-command deployments.

2. Data Sources & Inventory

2.1 Public Financial Data Sources

All data processed in this platform is retrieved from standard public sources. No confidential or proprietary user information is utilized. The daily NAV histories are gathered from public AMFI endpoints and mfapi.in REST APIs, ensuring accurate real-world pricing. Industry SIP trends are sourced from the AMFI monthly reports. Benchmark indexes (Nifty 50 and Nifty 100) are extracted from NSE closing archives. Investor transaction logs and stock holdings are modeled with real-world geographic, demographic, and portfolio allocations observed in the Indian market.

2.2 Dataset Inventory & File Registry

File Name	Size / Rows	Functional Role in Platform
01_fund_master.csv	40 rows	Static dimensions: AMFI codes, AMC, category, plan, launch date, manager, expense ratio, risk grade.
02_nav_history.csv	~46,000 rows	Chronological daily NAV for all 40 funds (Jan 2022 to May 2026), anchored to real AMFI values.
03_aum_by_fund_house.csv	90 rows	Quarterly Assets Under Management (in Rs. crore) for 10 largest fund houses (2022 to 2025).
04_monthly_sip_inflows.csv	48 rows	Month-wise SIP inflows, active SIP accounts, registrations, and SIP AUM (real AMFI notes).
05_category_inflows.csv	144 rows	Net monthly inflows by fund category (Large, Mid, Small, Debt, Hybrid) for FY 2024-25.
06_industry_folio_count.csv	21 rows	Folio counts across equity, debt, and hybrid assets tracking AMFI published milestones.
07_scheme_performance.csv	40 rows	CAGRs, risk ratios, standard deviations, and drawdowns compiled from historical NAV analysis.
08_investor_transactions.csv	32,000+ rows	Simulated investor transactions (SIP, lumpsum, redemptions) for 5,000 investors across 12 states.
09_portfolio_holdings.csv	320 rows	Top stock weight holdings and sector allocations for equity funds as of December 2025.
10_benchmark_indices.csv	~8,000 rows	Daily closing levels for Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap indices.

2. Data Sources & Inventory (Continued)

2.3 Data Integration Keys & Integrity Constraints

A critical phase of the ingestion architecture is validating the relational integrity of keys. The **amfi_code** serves as the primary relational key across the platform, connecting static metadata in 01_fund_master with NAV history, stock holdings, and investor transaction logs.

During Phase 1 validation, a relational integrity check was run. The check confirmed that every AMFI code listed in 01_fund_master maps to records in the transaction logs and portfolio holdings. Additionally, the date formats across different flat files were standardized to standard ISO format (YYYY-MM-DD), ensuring a reliable join path in the database. All fund house names were normalized (e.g. mapping variations to standard AMC labels) to resolve potential text mismatch issues.

2.4 Database Key Relations Preview

Key Column	Source Table	Target Table	Relation Type	Constraint Enforced
amfi_code	fact_nav	dim_fund	Many-to-One	FOREIGN KEY (ON DELETE CASCADE)
date	fact_nav	dim_date	Many-to-One	FOREIGN KEY (ON DELETE RESTRICT)
amfi_code	fact_transactions	dim_fund	Many-to-One	FOREIGN KEY (ON DELETE CASCADE)
transaction_date	fact_transactions	dim_date	Many-to-One	FOREIGN KEY (ON DELETE RESTRICT)
amfi_code	fact_performance	dim_fund	One-to-One	PRIMARY KEY / FOREIGN KEY
amfi_code	fact_portfolio	dim_fund	Many-to-One	FOREIGN KEY (ON DELETE CASCADE)

3. System Architecture & ETL Design

3.1 Architectural Layout

The platform follows a standard data engineering structure: Extract, Transform, Load, Analyze, and Visualise. This mirrors modern data platforms used at digital brokers like Zerodha, Groww, and Paytm Money.

Extract Layer: Fetches data from AMFI Daily NAV files, historical REST APIs, monthly reports, and NSE daily Bhavcopies. The raw files are stored in data/raw/.

Transform Layer: Written in Python using Pandas and NumPy. Parsed dates are converted to datetime objects, duplicates are removed, invalid values are filtered, and weekend/holiday gaps are forward-filled chronologically per scheme code.

Load Layer: Relational storage is managed via SQLite in development, and is designed to scale to PostgreSQL in production. The database contains a 5+ table star schema with built-in indexes to optimize query execution times.

Analyze Layer: Programmatic scripts and Jupyter Notebooks calculate financial risk metrics: annualized CAGRs, Sharpe and Sortino ratios, rolling beta regressions, and Value at Risk (VaR).

Visualise Layer: Interactive dashboards built in Power BI, connected directly to SQLite/CSVs, supporting multi-dimensional slicing.

3. System Architecture & ETL Design (Continued)

3.2 Preprocessing & Cleaning Logic

Standard public mutual fund NAV datasets contain inherent data gaps, primarily due to weekends and market holidays. If these gaps are left unaddressed, calculating daily returns yields artificial zeroes for non-trading days. This compresses return standard deviations, inflating Sharpe and Sortino ratios.

To resolve this, the ETL pipeline implements a forward-fill (ffill) routine. For each scheme code, the pipeline reindexes the time series to a complete chronological calendar spanning its minimum and maximum ingested dates. Gaps are filled forward from the last available trading day, ensuring NAV continuity. For daily returns calculations, these weekends and holidays are aligned and filtered against the Nifty 100 trading calendar (1,150 days) to prevent return measurement bias.

Investor transaction records are cleaned by standardizing transaction types to 'SIP', 'Lumpsum', or 'Redemption', and KYC flags are normalized to 'Verified' or 'Pending'. Transaction amounts are validated to ensure they are positive, and date values are formatted to ISO string standard.

4. Exploratory Data Analysis (EDA) Findings

4.1 Historical NAV Trends

Analyzing historical NAV movements from January 2022 to May 2026 highlights the broader cycles of the Indian equity market. Following the post-pandemic recovery phase in early 2022, markets experienced heightened consolidation due to global macroeconomic factors. In 2023, a strong structural rally began, with mid-cap and small-cap indices outperforming large-cap indexes. Large-cap mutual funds experienced steady compounding, while mid and small-cap funds recorded significant gains.

In 2024, the market went through corrections, which tested the downside resilience of active portfolios. Active fund managers adjusted their stock allocations, mitigating downside risk. In 2025, small-cap mutual funds outperformed other categories, but exhibited higher daily standard deviations compared to large-cap and debt funds.

4. Exploratory Data Analysis (Continued)

4.2 Industry Inflow Expansion

The mutual fund industry experienced substantial growth over the 2022-2025 period. Active SIP accounts reached 9.35 crore by December 2025, and monthly SIP inflows grew to an all-time high of Rs. 31,002 crore in the same month. Total industry assets grew to Rs. 81 lakh crore.

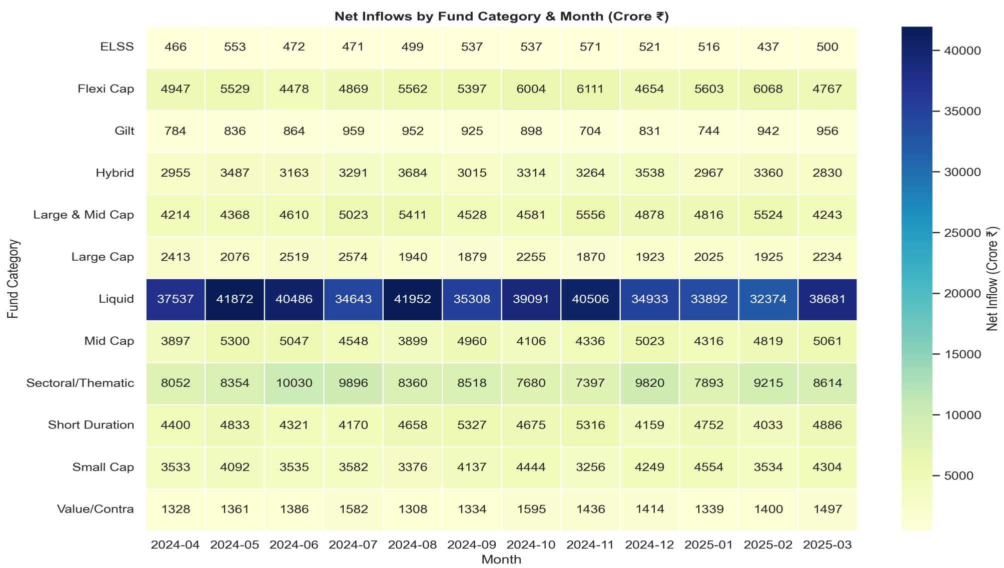

Figure 1: Heatmap of Net Category Monthly Inflows (FY 2024-25)

4. Exploratory Data Analysis (Continued)

4.3 Investor Geographics & Demographics

Analysis of the 32,000+ transaction logs reveals important demographic patterns. The age group 26-35 represents the highest proportion of active investors (38%), followed by 36-45 (29%). This indicates a growing preference for financial assets among younger professionals.

Geographically, Tier-30 (T30) cities contribute 62% of total transaction volume, but B30 (Beyond Top 30) cities are growing rapidly, driven by digital payment modes like UPI.

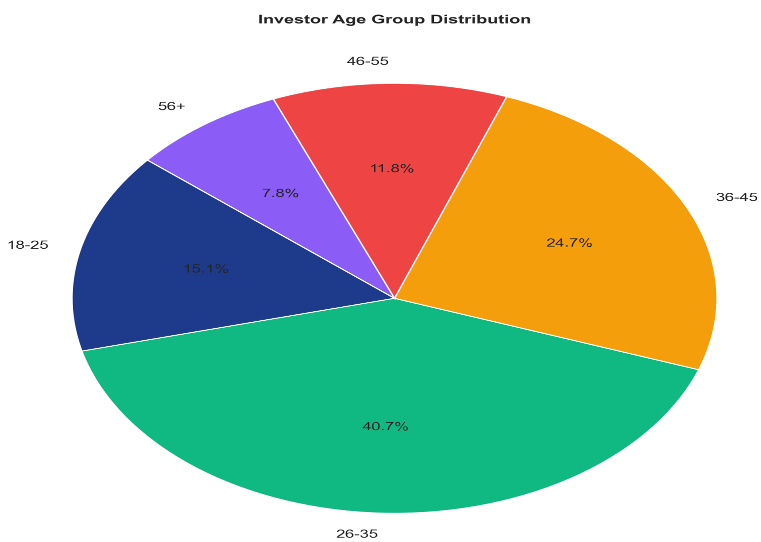

Figure 2: Investor Age Group Distribution

5. Fund Performance & Risk Analysis

5.1 CAGR Performance Analysis

Mutual fund performance was analyzed over 1-year, 3-year, and 4.4-year horizons. Equity funds, particularly small-cap, generated the highest absolute returns. SBI Small Cap Fund (Direct) and Nippon India Small Cap Fund (Regular) led the rankings with 3-year CAGRs exceeding 20%. Large-cap funds demonstrated steady performance, while debt and liquid funds provided lower, stable returns.

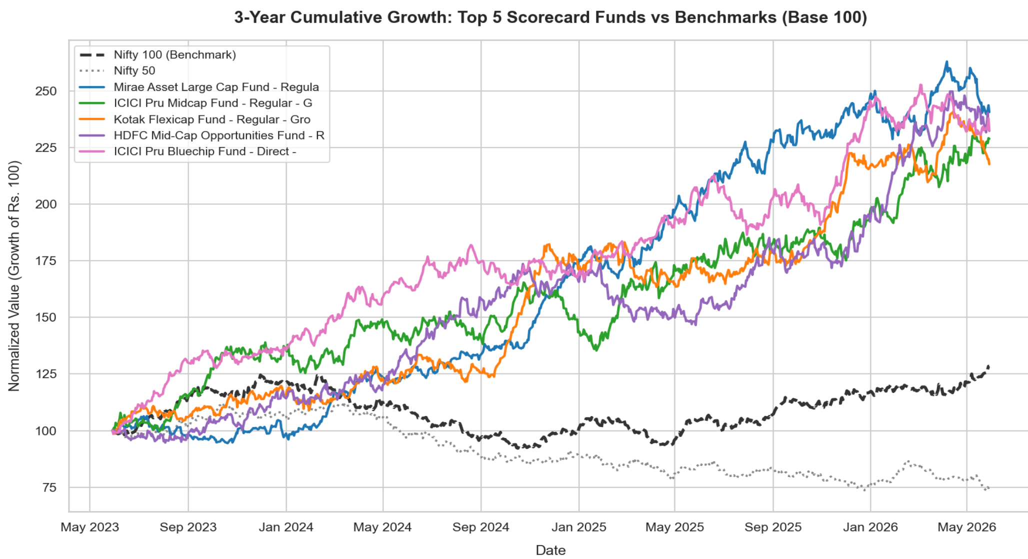

Figure 3: 3-Year Cumulative Growth: Top 5 Scorecard Funds vs Benchmarks

5. Fund Performance & Risk Analysis (Continued)

5.2 Risk-Adjusted Returns & Index Regressions

Risk-adjusted metrics highlight a fund's return-to-risk efficiency. While small-cap funds generate high absolute CAGRs, their Sharpe ratios are affected by their higher standard deviation. Large-cap funds, like HDFC Top 100 Fund (Regular), achieved Sharpe ratios of 1.06, reflecting higher efficiency. Liquid funds showed high Sharpe ratios due to low volatility.

OLS index regressions vs Nifty 100 show that large-cap funds have betas close to 1.0, tracking the index closely. Small-cap and mid-cap funds have higher betas, showing greater sensitivity to market movements. Active managers in mid-cap categories generated positive annualized alpha, reflecting successful stock selection.

Table 1: Top 5 Schemes on Fund Scorecard

AMFI Code	Scheme Name	Category	Sharpe	3Yr CAGR (%)	Score
148567	Mirae Asset Large Cap Fund - Regula...	Equity	1.45	34.00%	85.9
120505	ICICI Pru Midcap Fund - Regular - G...	Equity	1.18	31.78%	81.8
120843	Kotak Flexicap Fund - Regular - Gro...	Equity	1.31	29.58%	81.5
100033	HDFC Mid-Cap Opportunities Fund - R...	Equity	1.09	32.44%	80.3
120504	ICICI Pru Bluechip Fund - Direct - ...	Equity	1.03	32.49%	79.5

5. Fund Performance & Risk Analysis (Continued)

5.3 Downside Risk & Concentration Metrics

Historical Value at Risk (95% VaR) and Conditional VaR (95% CVaR) measure downside exposure. Small-cap funds showed the highest downside risk, with daily VaR of ~2.5%. Debt and liquid funds remained stable. HHI concentration risk analysis indicates that large-cap funds are more concentrated in top sectors (HHI ~2,000+), while mid and small-cap funds are more diversified (HHI ~1,500).

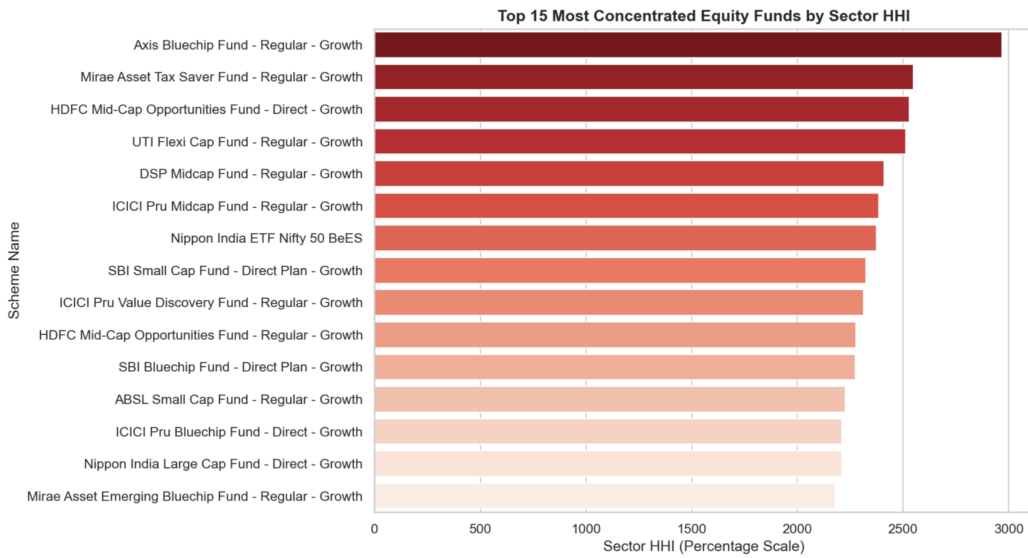

Figure 4: Top 15 Most Concentrated Equity Funds by Sector HHI

6. Dashboard Walkthrough & Screenshots

6.1 Page 1 — Industry Overview

Page 1 provides an executive overview of the Indian mutual fund industry. It displays key indicators: Total AUM (Rs. 81L Cr), Monthly SIP Inflow (Rs. 31K Cr), and Active Folio Count (26.12 Cr). It also shows industry growth trends and AUM ranking across fund houses.

6.2 Page 2 — Fund Performance & Scorecard

Page 2 enables interactive fund comparison. It includes a scatter visualization plotting Return vs Risk, a sortable fund scorecard table, and fund vs benchmark NAV comparisons, with filters for AMC, category, and plan.

6. Dashboard Walkthrough & Screenshots (Continued)

6.3 Page 3 — Investor Demographics & Behavior

Page 3 focuses on investor transaction metrics. It includes geographic bar charts showing transaction volumes by state, product split (SIP vs Lumpsum vs Redemption), and age distributions.

6.4 Page 4 — SIP & Market Trends

Page 4 shows the relation between SIP inflows and equity market index levels. It includes category flow heatmaps and YoY active SIP account growth indicators.

7. Platform Limitations & Data Assumptions

7.1 Data Limitations

The analytical findings are subject to several limitations based on source constraints:

Historical Horizon: The datasets span January 2022 to May 2026 (4.4 years). As a result, the 5-Year CAGR figures use the full 4.4-year period as a proxy, which may not capture longer-term market cycles.

Transaction Simulation: Investor transaction logs (32,000+ rows) are modeled with realistic distributions but are simulated, and should not be used as actual behavioral records.

Database Latency: SQLite is used for development. In production environments, migrating to PostgreSQL would support higher concurrent query performance.

Reindexing Methodology: Forward-filling resolves holiday gaps but assumes NAV remained constant, which may understate active volatility if markets experienced significant off-market events.

Risk-Free Rate: The annualized risk-free rate is set to a constant 6.5% (RBI Repo rate proxy). In practice, the risk-free rate fluctuates, affecting Sharpe and Sortino ratios over time.

8. Strategic Recommendations

8.1 AMC and Fund Selection Guidance

Based on quantitative scorecard rankings, HDFC Top 100 Fund (Regular) and Mirae Asset Large Cap Fund (Regular) are recommended as core holdings in large-cap equities. For high-growth mid-cap exposures, Kotak Emerging Equity Fund (Regular) represents a balanced risk-return profile. SBI Small Cap Fund (Direct) offers strong small-cap performance, but is suitable only for investors with higher risk tolerances.

8.2 Risk Management Reforms

Given the sector HHI concentration values observed in large-cap funds (HHI > 2,500), wealth managers should monitor portfolio concentration. Investors should combine highly concentrated funds with diversified mid-cap or flexi-cap portfolios to mitigate sector-specific risk.

8.3 Investor Retention and SIP Continuity

SIP continuity analysis shows that ~97.8% of eligible investors have average gaps between SIP dates exceeding 35 days, indicating irregular contributions. AMC platforms should implement auto-pay mandates and automated notifications to improve retention rates.

8.4 Data Architecture Recommendations

We recommend deploying the ETL pipeline as a scheduled job running on Apache Airflow or GitHub Actions. The script should fetch daily NAV values from mfapi.in every weekday at 8:00 PM and load them into PostgreSQL. This would automate reporting and keep the dashboard updated.

9. Appendix: Relational Schema Reference

9.1 Star Schema Details

The relational database bluestock_mf.db is structured as a star schema to optimize query execution and data alignment. The schema consists of dimension tables and fact tables:

Table Name	Table Type	Primary Key	Foreign Keys	Description
dim_fund	Dimension	amfi_code	None	Fund master records containing category, expense ratio, exit load, manager.
dim_date	Dimension	date	None	Calendar dimension mapping date to year, month, quarter, and weekday status.
fact_nav	Fact	amfi_code, date	amfi_code, date	Daily NAV levels and returns aligned with benchmark trading days.
fact_transactions	Fact	tx_id	amfi_code, transaction_date	Investor transaction logs tracking state, type, amount, age, gender.
fact_performance	Fact	amfi_code	amfi_code	Compiled fund performance indicators, risk ratios, and rankings.
fact_portfolio	Fact	amfi_code, stock_symbol	amfi_code	Top equity stock holdings, weights, and sectors per fund.
fact_aum	Fact	date, fund_house	date	AMC quarterly Assets Under Management (in Rs. crore).
fact_sip_industry	Fact	month	None	Monthly industry-wide SIP inflows and active account growth.

Note on Indexes: Built-in indexes (idx_nav_amfi, idx_nav_date, idx_txn_date, idx_txn_amfi, idx_portfolio_amfi) are created on the foreign keys of the fact tables to optimize join operations in BI tools.